Lecture 4 v2:
OLS Implementation

25117 - Econometrics

Universitat Pompeu Fabra

October 9th, 2024

What we learned in the last lesson

- The population regression line, $\beta_0 + \beta_1 X$, is the mean of Y as a function of the value of X.
  - The slope, β_1, is the expected difference in Y between two observations with X values that differ by one unit.
  - The intercept, β_0, determines the level (or height) of the regression line.

- The population regression line can be estimated using sample observations (Y_i, X_i) with $i = 1, \ldots, n$ by ordinary least squares (OLS).
  - The OLS estimators of the regression intercept and slope are denoted $\hat{\beta}_0$ and $\hat{\beta}_1$.
  - The predicted value of Y given X is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$.

What we learned in the last lesson

- The R^2 and standard error of the regression (SER) are measures of how close the values of Y_i are to the estimated regression line.
  - The R^2 is between 0 and 1, indicating how much variance in our outcome is explained by the variance of our regressors.
  - The standard error of the regression estimates the standard deviation of the regression error.

- There are three key assumptions for estimating causal effects using the linear regression model:
  1. The regression errors, u_i, have a mean of 0, conditional on the regressors X_i;
  2. the sample observations are i.i.d. random draws from the population; and
  3. large outliers are unlikely

- If these assumptions hold, the OLS estimator $\hat{\beta}_1$ is
  1. an unbiased estimator of the causal effect β_1
  2. consistent
  3. normally distributed when the sample is large.

Estimation of the regression line

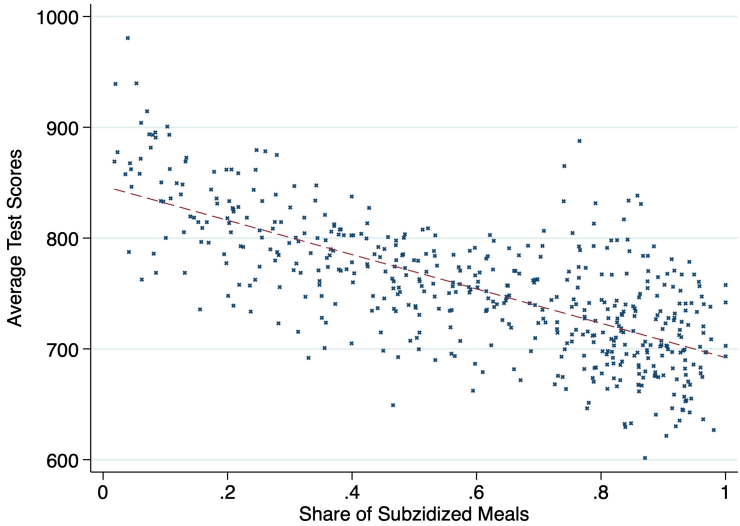
We want to learn about the slope of the population regression line. We have data from a sample, so there is sampling uncertainty. There are five steps toward this goal:

1. State the population of interest
2. Provide an estimator of this population object
3. Derive the sampling distribution of the estimator (this requires certain assumptions). In large samples, this sampling distribution will be normal by the CLT.
4. The square root of the estimated variance of the sampling distribution is the standard error (SE) of the estimator
5. Use the SE to construct t-statistics (for hypothesis tests) and confidence intervals.

Estimation of the regression line

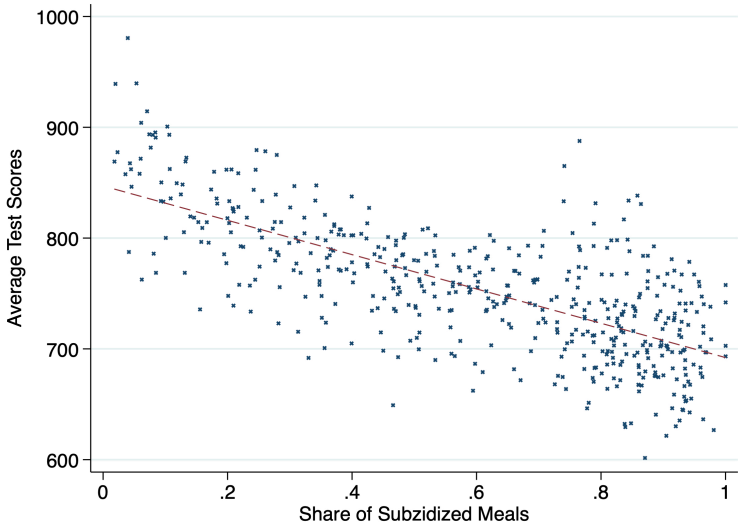
$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- β_1 is the slope of the population regression line
- $\hat{\beta}_1$ is the OLS estimator of β_1
- When n is large, by the Central Limit Theorem, the sampling distribution of $\hat{\beta}_1$ is well approximated by a normal distribution $\mathcal{N}(\beta_1; \frac{\sigma_u^2}{TSS_X})$



Hypothesis Testing

- Most of the time, we test for $H_0 : \beta_1 = 0$
- Null hypothesis and two-sided alternative:
  - $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 \neq \beta_{1,0}$
- Null hypothesis and one-sided alternative:
  - $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 < (or >)\beta_{1,0}$



Hypothesis Testing

- In general, the t-statistic is given by

$$t = \frac{\text{estimator} - \text{hypothetised value}}{\text{standard error of the estimator}}$$

- For μ_Y, the t-statistic is given by

$$t = \frac{\overline{Y} - \mu_{Y,0}}{s_Y/\sqrt{n}}$$

- For β_1, the t-statistic is given by

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)}$$

where $SE(\hat{\beta}_1) = \sqrt{\frac{SSR}{(n-2)TSS_X}} = \sqrt{\frac{\sum_{i=1}^{n}\hat{u}_i^2}{(n-2)\sum_{i=1}^{n}(X_i-\overline{X})}}$ when errors are homoskedastic (i.e., $var(u_i \mid X) = \sigma_u^2 \quad \forall i = 1, \ldots, n$)

- We reject $H_0 : \beta_1 = \beta_{1,0}$ at 5% significance level if $\mid t \mid > 1.96$, or alternatively, if $p - value < 0.05$

Stata Application

- We regress schools' average test scores (Y) against the share of subsidized meals (X).
- Our regression sample includes 500 schools ($i = 1, \ldots, 500$)

```
. reg testscore frpm_frac_s

      Source |       SS           df       MS      Number of obs   =       500
-------------+----------------------------------   F(1, 498)       =    501.86
       Model |  910811.722         1  910811.722   Prob > F        =    0.0000
    Residual |  903808.084       498  1814.87567   R-squared       =    0.5019
-------------+----------------------------------   Adj R-squared   =    0.5009
       Total |  1814619.81       499  3636.51264   Root MSE        =    42.601

------------------------------------------------------------------------------
   testscore | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  frpm_frac_s |  -154.8953   6.914289   -22.40   0.000    -168.4801   -141.3106
       _cons |    847.072   4.576515   185.09   0.000     838.0803    856.0636
------------------------------------------------------------------------------
```

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i$$

Stata Application

- The estimate of the intercept, $\hat{\beta}_0$ is equal to 847.072
- This is the value of the expected average test score for a school not subsidizing meals at all (i.e., $X_i = 0$)
- Because all schools in our sample data subsidize a non-zero share of meals, it does not really have a real-world interpretation

```
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```

$$Y_i = 847.072 + \hat{\beta}_1 X_i + \hat{u}_i$$

Stata Application

- The estimate of the regression slope, $\hat{\beta}_1$ is equal to -154.8953
- An 10%-pt increase in the share of subsidized meals is associated with a 15.49 decrease in the average level of test scores
- This represents a 25.69% reduction in the standard deviation of test scores

```
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------------------------------------------------------------------------------
```

$$Y_i = 847.02 - 154.8953 X_i + \hat{u}_i$$

Stata Application

- The estimate of the intercept, $\hat{\beta}_0$ is significantly different from 0.
- $\mid t_{\hat{\beta}_0} \mid = 185.09 >> 2.54$ (or, alternatively, $p - value < .01$
- We reject $H_0 : \beta_0 = 0$ at the 1% level

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```

$$Y_i = \underset{[4.576]}{847.02 - 154.8953 X_i} + \hat{u}_i$$

Stata Application

- The estimate of the regression slope, $\hat{\beta}_1$ is significantly different from 0.
- $| \ t_{\hat{\beta}_1} \ | = 22.4 >> 2.54$ (or, alternatively, $p - value < .01$
- We reject $H_0 : \beta_1 = 0$ at the 1% level

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```

$$Y_i = \underset{[4.576]}{847.02} \ \underset{[6.914]}{-154.8953 X_i} + \hat{u}_i$$

Stata Application

Recall that the CI defines

- The set of points that cannot be rejected at the 5% significance level;
- A set-valued function of the data (an interval that is a function of the data) that contains the true parameter value 95% of the time in repeated samples.

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```

$$CI_{\beta_0} = 847.072 + 1.96 \pm 4.576 = [838.0803; 856.0636] \notin 0$$
$$CI_{\beta_1} = -154.895 + 1.96 \pm 6.914 = [-168.4801; -141.3106] \notin 0$$

Stata Application

Remember that

- The SER measures the spread of the distribution of u around the regression line in the units of the dependent variable.
- It is an estimator of the standard deviation of the regression error u_i.

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```

$$SER = \sqrt{\frac{SSR}{n-2}} = \sqrt{\frac{\sum_{i=1}^{n}(\hat{u}_i - \bar{\hat{u}}_i)^2}{n-2}} = \sqrt{\frac{903808.084}{498}} \approx 42.601$$

Stata Application

Remember that

- The regression R^2 measures the fraction of the variance of Y that is explained by X
- It is unitless and ranges between zero (no fit) and one (perfect fit)

```
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```

$$R^2 = \frac{\text{explained sum of squares } (ESS)}{\text{total sum of squares } (TSS)} = \frac{\sum_{i=1}^{n}(\hat{Y}_i - \overline{Y})^2}{\sum_{i=1}^{n}(Y_i - \overline{Y})^2} = \frac{910811.722}{1814619.81} \approx 0.5019$$

Homoskedasticity vs. Heteroskedasticity

- In regression analysis, we often make an important assumption about the variance of the error term, u, known as **homoskedasticity**.
- Homoskedasticity implies that the variance of the error term is constant for all observations.
- However, in most cases, this assumption may not hold, leading to **heteroskedasticity**, where the variance of u varies across observations.

Homoskedasticity vs. Heteroskedasticity

For homoskedastic data, we assume:

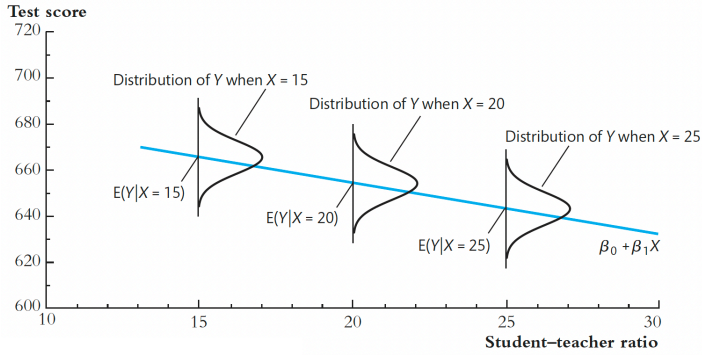
$$\text{var}(u_i \mid X) = \sigma^2 \quad \text{for all } i$$

For heteroskedastic data, the variance of u_i is not constant:

$$\text{var}(u_i \mid X) = \sigma_i^2 \quad \text{for each } i$$

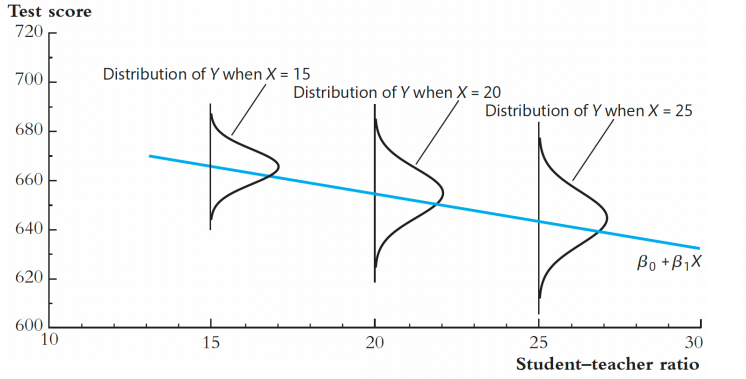
Graphical Illustration

- Understanding **homoskedasticity** and heteroskedasticity is crucial in regression analysis.
- **Homoskedasticity** assumes constant error variance, while heteroskedasticity implies varying error variance.
- To account for heteroskedasticity, robust standard errors can be used to obtain valid inferences.



Graphical Illustration

- Understanding homoskedasticity and **heteroskedasticity** is crucial in regression analysis.
- Homoskedasticity assumes constant error variance, while **heteroskedasticity** implies varying error variance.
- To account for **heteroskedasticity**, robust standard errors can be used to obtain valid inferences.



A conditional-expectation rule

When conditioning on X, any function of X is already known:

$$\mathbb{E}[XY \mid X] = X\mathbb{E}[Y \mid X].$$

Indeed, fixing $X = x$ turns x into a constant:

$$\mathbb{E}[XY \mid X = x] = \mathbb{E}[xY \mid X = x] = x\mathbb{E}[Y \mid X = x].$$

Because this holds for every possible value x, the random-variable version follows.

$var[\hat{\beta}_1]$ with robust standard errors

Start from the OLS estimation error and condition on X:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \overline{X}) u_i}{TSS_X}, \qquad TSS_X = \sum_{i=1}^n (X_i - \overline{X})^2.$$

Because X is fixed under the conditional variance, the weights are constants.

$$\begin{aligned}
var(\hat{\beta}_1 \mid X) = var(\hat{\beta}_1 - \beta_1 \mid X) &= var\left[\frac{\sum_{i=1}^n (X_i - \overline{X}) u_i}{\sum_{i=1}^n (X_i - \overline{X})^2} \middle| X \right] \\
&= var\left[\frac{\sum_{i=1}^n (X_i - \overline{X}) u_i}{TSS_X} \middle| X \right] \\
&= \frac{1}{TSS_X^2} var\left[\sum_{i=1}^n (X_i - \overline{X}) u_i \middle| X \right] \\
&= \frac{\sum_{i=1}^n (X_i - \overline{X})^2 var(u_i \mid X)}{TSS_X^2} \\
&= \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \overline{X})^2 var(u_i \mid X)}{\frac{n-1}{n} s_X^2}
\end{aligned}$$

The last equality uses independence across observations conditional on X.

$var[\hat{\beta}_1]$ with robust standard errors

The previous expression contains the unknown conditional variances $var(u_i \mid X)$. We estimate them using $\hat{u}_i^2$:

$$\sigma^2_{\hat{\beta}_1} = \frac{\sum_{i=1}^n (X_i - \mu_X)^2 var(u_i \mid X)}{n\sigma_X^2}$$

and $\hat{\beta}_1 \sim \mathcal{N}(\beta_1; \frac{\sigma_\nu^2}{n\sigma_X^2})$ where $\nu = (X_i - \mu_X)u_i$

The robust variance estimator is therefore:

$$\hat{\sigma}^2_{\hat{\beta}_1} = \frac{1}{n} \frac{\frac{1}{n-2}\sum_{i=1}^n (X_i - \overline{X})^2 \hat{u}_i^2}{\left[\frac{1}{n}\sum_{i=1}^n (X_i - \overline{X})\right]^2}$$

Finally, the robust standard error is $SE(\hat{\beta}_1) = \sqrt{\hat{\sigma}^2_{\hat{\beta}_1}}$.

In practice

- If the errors are either homoskedastic or heteroskedastic and you use heteroskedastic-robust standard errors, you are OK
- If the errors are heteroskedastic and you use the homoskedasticity-only formula for standard errors, your standard errors will be wrong

```
. reg testscore frpm_frac_s, robust

Linear regression                               Number of obs     =        500
                                                F(1, 498)         =     437.13
                                                Prob > F          =     0.0000
                                                R-squared         =     0.5019
                                                Root MSE          =     42.601

------------------------------------------------------------------------------
             |               Robust
  testscore  | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  frpm_frac_s|  -154.8953   7.40853    -20.91   0.000    -169.4512   -140.3395
       _cons |   847.072    4.807364   176.20   0.000     837.6268    856.5172
------------------------------------------------------------------------------
```

- The homoskedasticity-only estimator of the variance of $\hat{\beta}_1$ is inconsistent if there is heteroskedasticity.

The Theoretical Foundations of OLS

Let us consider a linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Subject to the following assumptions:

1. Zero Conditional Mean: The errors u_i have a conditional mean of zero given the values of the independent variables, i.e., $\mathbb{E}(u_i|X_i) = 0$ for all i.
2. Random Draws: observations $(Y_i; X_i)$ are randomly drawn from the same population distribution, i.e., they are i.i.d. for all i.
3. Large outliers are rare: $\mathbb{E}(Y^4) < \infty$ and $\mathbb{E}(X^4) < \infty$
4. Homoskedasticity: The errors have constant variance, i.e., $\text{Var}(u_i) = \sigma_u^2$ for all i.
5. u is normally distributed: $u \sim \mathcal{N}(0; \sigma_u^2)$.

Gauss-Markov Theorem

Under assumptions 1-4, the OLS estimators are Best Linear Unbiased Estimators (BLUE).

The Gauss-Markov Theorem

- The OLS estimator $\hat{\beta}_1$ is a linear estimator, that is, it can be written as a linear function of $Y_1, ..., Y_n$:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})u_i}{\sum_{i=1}^{n}(X_i - \overline{X})^2} \equiv w_i^{OLS} u_i$$

$$\text{where} \qquad w_i^{OLS} = \frac{\sum_{i=1}^{n}(X_i - \overline{X})}{\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

- The Gauss-Markov theorem states that among all possible choices of $\{w_i\}$ the OLS weights w_i^{OLS} yield the smallest $var(\hat{\beta}_1)$.
- Under assumptions 1-5, the ordinary least squares (OLS) estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ have the minimum variance among all (linear and non-linear) unbiased estimators

The Gauss-Markov Theorem: Limits

The Gauss-Markov Theorem is a crucial justification for using OLS in many regression analyses, but it is very restrictive and, therefore, has serious practical limitations.

- The result is only for linear estimators –- only a small subset of estimators
- The condition of homoskedasticity often doesn't hold (homoskedasticity is a special case)
- When estimating the population mean, if there are significant outliers, the median is preferred over the mean because it's less sensitive and has lower variance. Similarly, in regression, OLS estimators can also be sensitive to outliers. In such cases, other estimators like the Least Absolute Deviations (LAD) estimator may be more efficient:

$$\min_{\hat{\beta}_0;\hat{\beta}_1} \sum_{i=1}^{n} \mid Y_i - \beta_0 - \beta_1 X_i \mid$$

Back to the original question

Suppose schools receive extra transfers from the central government so the share of subsidized school meals increases by 10%-pt. What is the effect of this policy intervention ("treatment") on test scores?

- This question requires an estimate of the causal effect on test scores of a change in the share of subsidized meals. Does our regression analysis using the California data set provide a compelling estimate of this causal effect?
- Not really – districts with a low share of subsidized meals tend to be ones with lots of other resources and higher income families, which provide kids with more learning opportunities outside school...this suggests that $corr(u_i; X_i) > 0$ (that is, $\mathbb{E}(u_i \mid X_i) \neq 0$).
- It seems that we have omitted some factors, or variables, from our analysis, and this has biased our results...

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 5